

Unit 5 Vocabulary



What would you do if...?

Inglés Profesional





Table of Contents

5.1. Home automation.Domotics

- 5.1.1. Main Benefits of Home Automation
- 5.1.2. Types of Home Automation Systems.
- 5.1.3. Key components of a home automation system
- 5.1.4. Levels of home automation.

5.2. The electricity bill

- 5.2.1. Main Parts of the Electricity Bill
- 5.2.2. Other common charges
- 5.2.3. Back Side of the Bill: Detailed Charges
- 5.2.4. Electricity tariffs

5.3. Let's practise!

5.4. Test yourself!

5.5. Answer key

Test yourself - Answer key



5.1.

Home automation. Domotics

Domotics, also known as **home automation**, refers to the integration of technology and services within a home environment to enhance comfort, efficiency, and security. The word *domotics* is derived from the Latin word "**domus**" (meaning home) and the terms **informatics**, **telematics**, and **robotics**.

Home automation allows the **automatic control** of household devices and systems, such as **lighting**, **heating**, **air conditioning**, **security systems**, and **appliances**. These systems are often connected to the **Internet**, enabling **remote control** through devices such as smartphones, tablets, or voice assistants.

5.1.1. Main Benefits of Home Automation

- > **Convenience**: automating daily tasks such as turning off lights or locking doors.
- > **Energy efficiency**: smart thermostats and lighting systems help reduce electricity consumption.
- > **Security**: motion sensors, smart locks, and surveillance cameras offer improved safety.
- > **Cost savings**: by optimizing energy use, users save on heating, cooling, and electricity bills.

5.1.2. Types of Home Automation Systems

Depending on the kind of controls and connections there are three types of home automation systems.

- > **Power Line Home Automation System**. These systems use existing electrical power lines to transfer data and commands.
 - » **Advantage**: no need for extra wiring.
 - » **Disadvantage**: complex setup requiring **converter circuits** and special **interfaces**.
- > **Wired Home Automation System**: In this type, all devices are connected through cables to a **central controller**.
 - » **Advantage**: high reliability and low interference
 - » **Disadvantage**: installation is more expensive and invasive due to cabling.
- > **Wireless Home Automation**: These use technologies like **Wi-Fi**, **ZigBee**, or **Bluetooth** to connect devices.
 - » **Advantage**: easy to install and expand.
 - » **Disadvantage**: it may suffer from signal interference.



5.1.3. Key components of a home automation system

- > **Controllers:** devices (like smartphones, tablets, or hubs) that manage the behaviour of smart systems.
- > **Sensors:** devices that detect environmental changes (e.g., motion, temperature, light).
- > **Actuators:** devices that perform physical actions, such as opening a valve or turning off lights.
- > **HMI (Human-machine interface):** touchscreens, apps, or voice assistants that allow users to interact with the system

5.1.4. Levels of home automation

1. **Basic Smart Homes:** they contain intelligent items (e.g., smart bulbs or thermostats).
2. **Communicative Homes:** devices that communicate with each other to improve coordination.
3. **Connected Homes:** they include internal and external networks for interactive control and remote access.
4. **Learning Homes:** these store data and patterns to predict user behaviour.
5. **Attentive Homes:** they monitor activity and location in real time to anticipate and meet the occupant's needs.





5.2.

The electricity bill

An **electricity bill** provides detailed information about how much electric energy a household has used over a certain period (usually one month). Electricity is measured in **kilowatt-hours (kWh)**, and 1 kWh is equivalent to using 1,000 watts for one hour.

5.2.1. Main Parts of the Electricity Bill

- > **Supply Service:** The cost of the electricity you consumed.
- > **Delivery Service:** The cost of transporting electricity through the power grid.
- > **Standard Supply Service Charge:** A fixed administrative fee.

5.2.2. Other common charges

- > **Meter Multiplier:** Adjusts meter readings when usage is too high to be displayed directly.
- > **Metering Charge:** Cost for reading and maintaining your electric meter.
- > **Customer Charge:** Covers customer support and account services.
- > **Distribution Charge:** Relates to the infrastructure used to deliver electricity to your home.

5.2.3. Back Side of the Bill: Detailed Charges

- > **Average Cost per Day:** your daily expense based on average usage.
- > **Average Daily Usage:** the amount of electricity consumed per day.
- > **Charges for Peak and Off-Peak Use:** some tariffs offer different prices depending on the time of day.

5.2.4. Electricity tariffs

- > **Flat Rate:** A fixed price per kWh throughout the day.
- > **Time-of-Use Tariff:** Different prices depending on the time electricity is used.
- > **Flexible Plan:** Combines different rates, with lower prices during **off-peak** hours and higher during **peak** hours.



5.3.

Let's practise!

1. Label the elements of a SMART HOUSE.

Socket	Light switch	Video door phone system	
Camera	Smart windows	Intercom	Master panel



1.



2.



3.



4.



5.



6.

2. Complete the sentences with the correct expression.

Standard Supply Service	Delivery Service
kilowatt-hours (kWh)	Time-of-Use Tariff

- The amount of electricity used in a home is measured in _____.
- A _____ covers the cost of transporting electricity to your home.
- The _____ charge is a fixed administrative fee included in the bill.
- A _____ plan charges different rates depending on the time of day.



5.3.

Test yourself!

1. What does the term “domotics” mean?
 - a. Domestic cleaning systems
 - b. Automatic kitchen appliances
 - c. Home robotics combining technology and services
 - d. Entertainment systems at home
2. Which of the following is a benefit of home automation?
 - a. Higher energy consumption
 - b. Increased electricity cost
 - c. Manual control of all devices
 - d. Improved comfort and efficiency
3. Which system uses existing electrical wiring to transmit data?
 - a. Wired home automation
 - b. Power Line home automation
 - c. Wireless system
 - d. Manual system
4. What is the role of actuators in a smart home?
 - a. Monitoring light and temperature
 - b. Operating mechanisms like motors or switches
 - c. Connecting the internet
 - d. Sending voice commands
5. Which of these devices is considered a controller?
 - a. Smartphone
 - b. Motion sensor
 - c. Refrigerator
 - d. Smart valve
6. What is a characteristic of a “learning home”?
 - a. Controls are only manual
 - b. It adjusts automatically based on user behaviour
 - c. It has no internet connection
 - d. It only supports basic lighting



7. What does the electricity bill measure?
 - a. Gas consumption
 - b. Temperature changes
 - c. Water usage
 - d. Energy use in kilowatt-hours
8. What is a 'meter multiplier' used for?
 - a. To convert meter readings for large usage
 - b. To clean electric meters
 - c. To reduce electricity costs
 - d. To reset daily usage
9. Which tariff charges the same price throughout the day?
 - a. Time-of-use
 - b. Peak tariff
 - c. Flexible plan
 - d. Flat rate
10. What does "off-peak" mean in terms of electricity use?
 - a. High-demand time
 - b. Night hours with reduced rates
 - c. Midday peak pricing
 - d. System shutdown time





5.4.

Answer key

Let's practise!

1.

1. Intercom
2. Light switch
3. Master panel
4. Smart windows
5. Video door phone system
6. Camera

2.

1. kilowatt-hours (kWh)
2. Delivery Service
3. Standard Supply Service
4. Time-of-Use Tariff

Test yourself - Answer key

- | | | |
|------|------|-------|
| 1. C | 5. A | 9. D |
| 2. D | 6. B | 10. B |
| 3. B | 7. D | |
| 4. B | 8. A | |



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